



Near real-time mapping of air temperature at high spatiotemporal resolutions in Tasmania, Australia

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Abstract

Surface air temperature (T_a) is important for a wide variety of applications that require real-time monitoring of environmental change; yet, few studies have attempted to deliver such products at a resolution appropriate for local-scale analysis. We investigated the feasibility of producing hourly T_a maps in near real time at a spatial grid resolution of 80 m across the state of Tasmania (area: 68,401 km²), Australia. We first assimilate 267 non-telemetered logger recording sites for data recorded in the 1-year period from September 2013 to 2014 and statistically calibrate them to 43-telemetered Australian Bureau of Meteorology (BoM) weather station sites for real-time application. This was evaluated in the following year using held back recordings in addition to evaluating regression trees (RT) and thin plate splines (TPS) to interpolate the hourly T_a estimates. For real-time operational mapping, the system was fully automated in the R programming language and hosted on a cloud-based computing platform to assess performance over a 7-day period in February 2020. The calibration procedure yielded accurate results with the root mean square error (RMSE) ranging between 1.33 °C in summer and 1.29 °C in winter. The TPS interpolation method was optimal in summer with an RMSE of 1.35 °C compared to RT with 1.39 °C; however, RT performed better in winter with an RMSE of 1.34 °C compared to 1.42 °C. The mapping system was capable of producing spatial outputs within the hour of the BoM observations becoming available with the TPS interpolation proving to be more efficient at producing outputs in a timelier manner.

1 Introduction

Air temperature (T_a) recorded within a few metres of the earth's surface is an important variable for weather and climate studies and critical input to many environmental modelling applications (Lazzarini et al. 2014; Wang and Zeng 2013; Xu et al. 2018). Furthermore, its derivation in a timely and efficient manner at a high spatial and temporal resolution is becoming increasingly important for a wide variety of applications requiring real-time monitoring of environmental change as well as input to numerical weather model data assimilation streams (Cheng et al. 2016; Williams et al. 2011). Such data are often available from a network of telemetered meteorological weather stations that stream live data to a server for broader use. However, such data tend to be generated from sparsely or irregularly located stations that fail

to take into account the spatial variability of temperature that can vary over short distances (Lazzarini et al. 2014; Webb et al. 2016). This can be problematic for locations that rely on observations from meteorological stations situated kilometres away from their location of interest, thereby providing an inaccurate representation of the true temperature (Jeffrey et al. 2001; Liu et al. 2018). In these circumstances, such inaccuracies can be highly influenced by local topographic factors where T_a can vary greatly, particularly in highly undulating and mountainous terrain (Dodson and Marks 1997; Hutchinson 1991; Jarvis and Stuart 2001a; Rolland 2003). This can be further complicated by coastal, urban and latitudinal effects (Jarvis and Stuart 2001a; Legates and Willmott 1990; Wang et al. 2011). It is therefore important that the real-time monitoring of T_a be determined spatially but also at an appropriate resolution to account for variability associated with various environmental factors.

Much research has been conducted that spatially quantify T_a using various interpolation techniques at various spatial resolutions (Dodson and Marks 1997; Hutchinson 1991; Jarvis and Stuart 2001b; Jeffrey et al. 2001; Jones et al. 2009; Webb et al. 2016; Xu et al. 2018). However, its application for real-time monitoring purposes, particularly at high spatiotemporal resolutions, are limited. A study by Lazzarini

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et al. (2014) used remote sensing and in situ meteorological station data to produce near real-time T_a maps across the United Arab Emirates. This was performed at sub-hourly temporal resolution using a land surface temperature model generated by combining satellite imagery with meteorological recordings linked to the seasons versus the time of day. Evaluation of the model showed that it was capable of producing predictions with a root mean square error 2.44 °C. The method was largely applied to a non-heterogeneous arid environment at a spatial resolution of ~ 3 km, thus limited in its ability to capture the subtleties associated with highly variable terrain. A study by Aadhar and Mishra (2017) produced a dataset based on precipitation and land surface temperature capable of monitoring drought conditions in near real time over South Asia. Daily minimum and maximum air temperature were produced from the Global Ensemble Forecast System (Hamill et al. 2013) that was bias-corrected and down-scaled to a spatial resolution of 0.05° (~ 5 km). Although a useful tool, the temporal and spatial resolution would warrant significant modification for high-resolution monitoring at the local scale. Similarly, the Scientific Information for Land Owners (SILO) database and the Australian Gridded Climate Data (AGCD) used trivariate thin plate smoothing splines (TPS) and a topography-resolving method to produce spatial grid estimates of daily minimum ($T_{\min}$) and maximum ($T_{\max}$) temperatures across Australia (Jeffrey et al. 2001; Jones et al. 2009). Both systems are updated daily (time lag of 1 day) using recordings garnered from Australian Bureau of Meteorology (BoM) weather stations with interpolated products presented at 0.05° (~ 5 km) grid resolution. Evaluation of the SILO system showed it was capable of predicting daily maximum and minimum temperatures with root mean square errors of 1.5 °C and 1.9 °C, respectively, whereas the AGCD data had similar errors of 1.2 °C and 1.7 °C. The SILO predictions can be readily accessed via the web (www.longpaddock.qld.gov.au/silo) and used to supplement purpose-built applications to monitor climate variability on a daily basis as demonstrated by the Australian CliMate App (Australian CliMate Development Team 2016). In a similar vein, Webb et al. (2016) used BoM weather station recordings statistically linked to short-term temperature logger recordings (174 in total) to produce long-term calibrated climate data for a study area (60,000 ha) in Tasmania, Australia. These were used for regression tree (RT) interpolation using terrain covariates to guide predictions of high-resolution (30-m grid spacing) temperature-based climate maps concerning frost risk, chill hours and growing degree days. The maps were used to identify new sites suitable for enterprise horticulture (Kidd et al. 2015) including new sites for viticulture (Webb et al. 2018). Although used for climate mapping applications, the methodology has the potential to contribute and enhance real-time spatial monitoring of T_a at a spatial resolution suitable for local-scale applications.

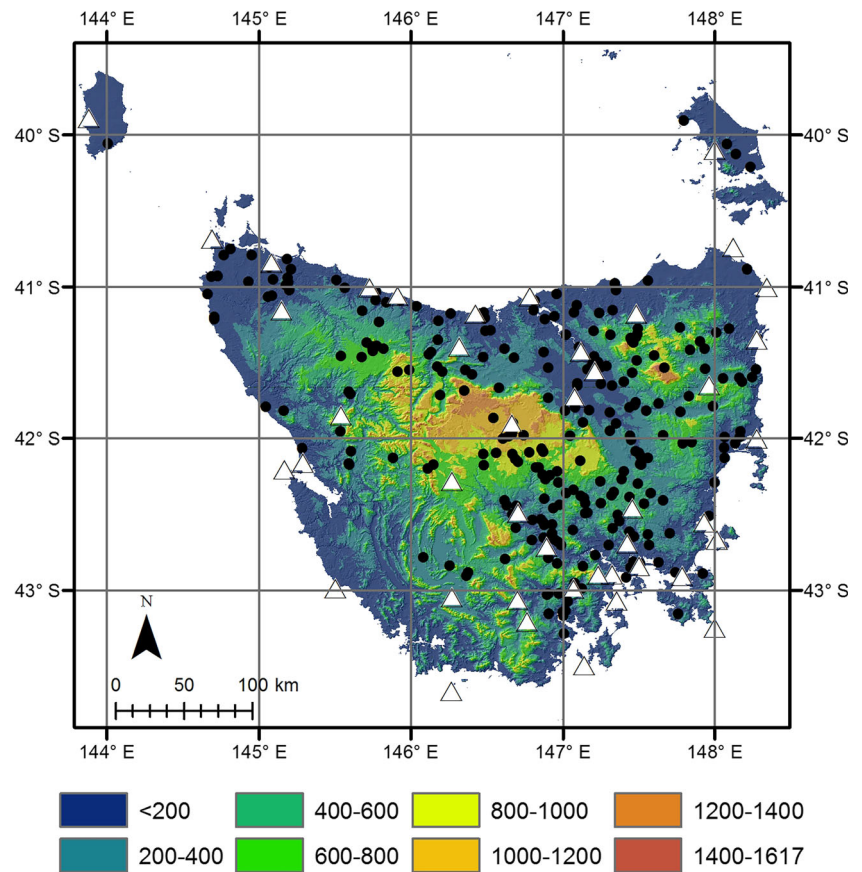
This study aimed to further develop the methods outlined in Webb et al. (2016); Webb et al. (2018) and Jeffrey et al. (2001) for operational real-time spatial monitoring of T_a at a high spatiotemporal resolution across the state of Tasmania. We first adopt the method prescribed by Webb et al. (2016) to calibrate and statistically link historic T_a recordings provided by 267 temperature logger sites (non-telemetered) to a network of 43 BoM weather station sites (telemetered) using hourly T_a data captured over a 1-year period. We then assess the accuracy of the calibration method using independent logger records in the following year in addition to trialling two interpolation methods: RT as applied in Webb et al. (2016) and TPS as applied in Jeffrey et al. (2001). Both methods were evaluated for their accuracy in delineating T_a predictions across the state at a spatial resolution of 80 m and assessed for the purpose of operational near real-time mapping of T_a (delivered hourly). Very few studies have attempted to deliver such products at a high spatiotemporal resolution over a considerably large land area (68,401 km²). This study provides the framework for delivering such products without the need to rely on live recordings sourced from distantly located weather stations that are comparatively less accurate over space and time.

2 Methods

2.1 Study area

Tasmania is an island (68,401 km²) located at coordinates 42.08° S, 146.59° E, approximately 240 km south of mainland Australia. The climate is classified as cool temperate (Australian Bureau of Statistics 2018) with four distinct seasons with summer occurring in months December to February, autumn: March to May, winter: June to August and spring: September to November. The mean average air temperature for winter averages around 8 °C for coastal fringes to around 1 °C for the high elevation areas. In summer, average temperatures increase to 16 °C and 9 °C for coastal and high elevation areas, respectively. Rainfall is highest during the winter months with the total annual rainfall, exhibiting a trend that decreases from west (> 1800 mm/year) to east (< 450 mm/year) (Bureau of Meteorology 2015). The topography of the state (Fig. 1) is characterised by rugged, highly undulating terrain in the west and south-west with the central area of the state forming the central highlands with elevation averaging 1000 m above sea level. The highest peak of the state is 1617 m. Lands located in the central east form the midland agricultural area, dominated by flat homogenous terrain interspersed with small rolling hills. The eastern half of the state is a mixture of variable topography with small to medium-sized hills and mountains with farmland situated along the coastal fringes.

Fig. 1 Elevation (m) map of the study area with location of temperature logger (black dots) and telemetered Bureau of Meteorology (BoM) weather station sites (white triangles)



2.2 Approach

The workflow of the current study was split into three parts (Fig. 2). Firstly, retrospective analysis for the period 23 September 2013 to 22 September 2014 (i.e. calibration period) was carried out to incorporate historic logger recording sites into the analysis using the calibration procedure outlined in Webb et al. (2016) and Webb et al. (2018). This involved regressing hourly logger recordings (T_a) against hourly spatially interpolated T_a estimates (i.e. BoM grids)—formed from hourly BoM weather station recordings—to produce calibration equations pertaining to each logger site (refer to part 1, Fig. 2). The equations were then used to bias correct future BoM grid estimates made in the second year (23 September 2014 to 22 September 2015, i.e. the evaluation period) and assessed against actual logger observations held back from the calibration procedure (refer to part 2, Fig. 2). This was in addition to cross validation of the regression tree (RT) and TPS interpolation procedures—adopted to map the calibrated predictions across the state in a continuous manner at a spatial resolution of 80 m. Finally, the proposed interpolation methods were applied to the operational real-time mapping of hourly T_a over a 7-day period from 7 to 13 February 2020. This involved using the calibration equations (derived in part 1, Fig. 2) to bias correct live data feeds—as provided

by the BoM weather station network—followed by RT and TPS interpolation of the T_a predictions across the state (refer to part 3, Fig. 2). The system was fully automated in R programming language (R Development Core Team 2015) and hosted on a cloud computing allocation provided by the Australian National eResearch Collaboration Tools and Resources project (NeCTAR) where overall processing performance was assessed.

2.3 Air temperature data

Two type of datasets were considered for the analysis. Firstly, T_a data recorded by automatic weather stations from the BoM and capable of providing real-time access to hourly T_a recordings were assessed for use. The purpose of this data was to create BoM grids and contribute to the RT and TPS interpolation as illustrated in Fig. 2. A key criterion was that each station possesses historic recordings retrospective to September 2013 to assimilate historic logger recordings into the analysis, as shown in parts 1 and 2 in Fig. 2. This was in addition to having current hourly T_a recordings capable of providing real-time access via the Internet, i.e. to apply the proposed methodology in real time, as shown in part 3 in Fig. 2. In total, there were 43 stations (Fig. 1) that record temperature at various frequencies ranging from every minute to an

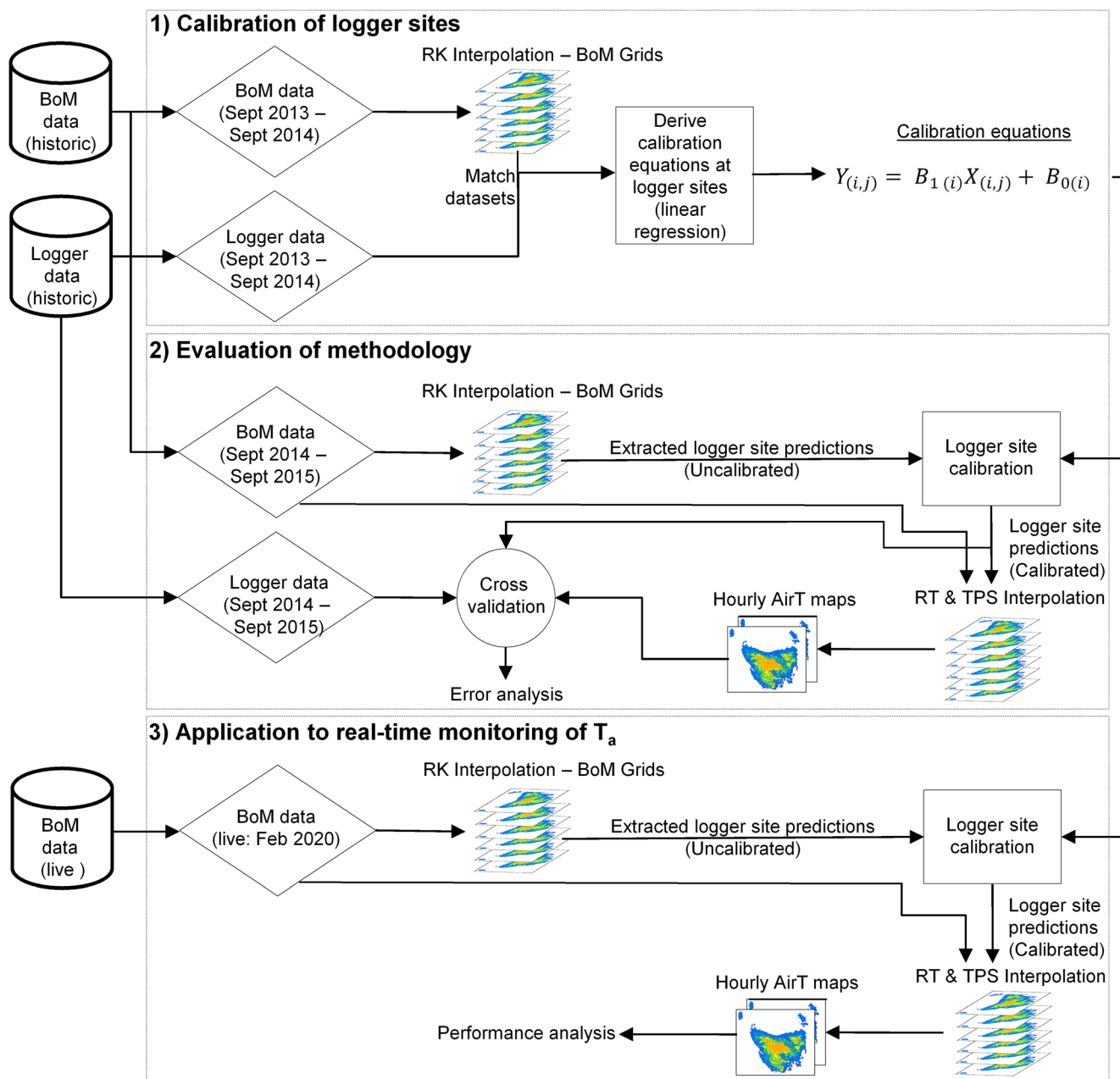


Fig. 2 Flow chart of the methodology developed in this study

hour. Observations occur 1.2 m aboveground using a resistance temperature detector housed in a Stevenson weather screen (Bureau of Meteorology 2018). All station recordings can be accessed ‘live’ via the BoM website (<http://www.bom.gov.au/tas/observations/>) and are typically available every half hour with a 10-min to 20-min delay from the observation recording time. Appraisal of the data dating back to September 2013 showed only minor recording gaps (< 1 year of missing recordings) rendering it satisfactory for the analysis.

The second type of dataset considered for the analysis and to complement the existing network of telemetered BoM station data was the acquisition of 267 non-telemetered historic

T_a recordings made available courtesy of the Tasmanian Government Department of Primary Industries, Parks, Water and Environment (DPIPWE) (Fig. 1). These are referred to as logger sites, where data were used to bias correct BoM grids for RT and TPS modelling, as shown in Fig. 2. The recordings were registered by Tinytag temperature loggers (model no. TGP 4017, Gemini Data Loggers, Chichester, England). These were strategically deployed around the state from June 2013 using a stratified sampling approach based on topographic variables, as advocated in Webb et al. (2016). Each logger was housed in a Datamate Weather Screen (model no. ACS-5050, Hastings Data Loggers, Port Macquarie, NSW,

Australia) and protected within a purpose-built steel cage with dimensions measuring $150 \times 30 \times 30$ cm. They were programmed to record T_a at 1.2 m aboveground level every 30 min from 23 September 2013 until 22 September 2015. The recordings were manually downloaded after this period and placed in a purpose-built database for querying. The recordings underwent quality control to remove erroneous data and/or missing values (due to battery failure). Loggers that did not meet the minimum requirement of continuous hourly recordings of at least 1 year were discarded from further analysis.

In total, the data screening process resulted in two separate datasets: one dataset comprising of live-telemetered hourly T_a recordings provided by 43 BoM stations dating back to September 2013 until the current day (available as real-time data feeds). Another logger dataset comprising of historic T_a data garnered from non-telemetered logger recordings for the period 22 September 2013 to 23 September 2015 (Fig. 2).

2.4 Calibrating logger sites for real-time T_a analysis

The calibration method outlined by Webb et al. (2016) was employed to assimilate recordings from logger sites with the BoM grids. The method first involves forming BoM grids using hourly T_a observations generated from BoM weather station sites across a period that is consistent with the logger recordings. In this case, hourly data between 23 September 2013 and 22 September 2014, i.e. the calibration period. Regression analysis between the grid estimates and the logger records was then conducted to relate the BoM grid predictions onto each logger site. The calibration procedure produces a set of linear equations that are applied to BoM grid estimates to predict T_a estimates at each of the logger sites.

To form the BoM grids, the regression kriging (RK) interpolation method (Odeh et al. 1995) was used along with covariate data to spatially de-trend the hourly BoM observations. The RK method has a deterministic and stochastic component. The deterministic component involves forming a regression between environmental variables and measured temperature, and the residual of the model is treated as a stochastic component that has a mean of zero and a spatial variability model of kriging. The main covariate used for the regression part of the interpolation process was the 3-arc sec (80-m resolution) digital elevation model (DEM) (Gallant et al. 2011). The DEM was also used to derive the following derivatives: the System for Automated Geoscientific Analyses (SAGA) wetness index (Böhner and Antonić 2009; Grabs et al. 2009), a distance to sea index (derived using the Euclidean Distance Spatial Analyst tool in ESRI 2013 ArcGIS Desktop: Release 10.2: Environmental Systems Research Institute, Redlands, CA, USA) and latitude/longitude grids. These covariates can account for a large proportion of T_a variability and were ideal for de-trending the hourly BoM data (Jarvis and Stuart 2001b; Webb et al. 2016). This was confirmed when the performance

of regression models—applied using software R (R Development Core Team 2015)—registered an average R^2 of 0.76 indicating that a significant amount of the T_a variability was accounted. The unaccounted variation of the hourly data, i.e. residual values, was then modelled via simple kriging to account for spatial autocorrelation. This procedure was done using the gstat module in R (Pebesma 2004) with variograms fitted automatically using an exponential model for each subsequent grid produced. The result of the RK interpolation process resulted in 8760 BoM grids produced individually for each hour within the calibration period.

Predictions from the RK were then spatially extracted at the geographic location of each logger site to provide an uncalibrated set of BoM grid predictions. Corresponding logger data and BoM grid estimates were then matched for each hour to undergo linear least-squares regression to produce an equation to describe the relationship of the BoM grids to the logger sites:

$$Y = B_1X + B_0 \quad (1)$$

where Y is the predicted (calibrated) temperature estimate for the logger site, X is the independent T_a BoM grid raster cell value (estimates extracted from the geographic location of the logger site), B_0 represents the intercept and B_1 represents the slope. The equation can subsequently be used to bias correct incoming BoM grid data generated from the telemetered BoM weather stations for hours outside of the calibration period, i.e. the evaluation period (23 September 2014 to 22 September 2015) and for real-time analysis (Fig. 2), in the following manner:

$$Y_{(i,j)} = B_1(i)X_{(i,j)} + B_0(i) \quad (2)$$

where $Y_{(i,j)}$ represent the bias-corrected T_a prediction, $X_{(i,j)}$ represent the T_a estimate extracted from the BoM grid raster at the logger site (i), and (j) corresponding to the hour outside the calibration period. Note that the calibration equations were derived separately for each season which was found to marginally improve the resulting calibrated predictions (Webb et al. 2016).

2.5 Interpolating T_a using regression trees and thin plate smoothing splines

Calibrated T_a estimates provided by the calibration procedure were interpolated on an hourly basis. This was done using regression trees (RT) and TPS to form continuous T_a maps in the evaluation period (23 September 2014 to 22 September 2015) as well as for application in real-time prediction (Fig. 2). The RT algorithm was chosen due to its robustness and ability to successfully map daily T_a , frost risk, chill hours and growing degree days to a high degree of accuracy and its relatively superior performance for producing predictions compared to other machine learning algorithms (Noi et al. 2017; Quinlan 1986; Webb et al. 2016; Webb et al. 2018; Xu et al. 2018). In its simplest form, the algorithm is a rule-based regression

technique and works by partitioning training data into groups of linear models to form a tree-like structure, similar to using piecewise linear functions (Quinlan 1992). To arrive at a prediction, the algorithm uses a splitting criterion based on computing the standard deviation of the target values from a set of training cases. This is a recursive process in order to evaluate every possible scenario relating to the training cases with the purpose of developing a training case that provides the greatest reduction in error. The result of the splitting process often results in large tree-like structures constituting subtrees grown on top of parent trees. This tends to result in overfitting of the training values, which is mitigated by ‘pruning’, or more specifically the use of multivariate linear models to replace the subtree structure. These models are based on attributes that were originally identified at the criterion splitting stage to maintain objectivity (Pal and Deswal 2009). The process is further simplified by removing parameters that contribute little to the overall model which serves to decrease overall model error. Thus, the final prediction is based on satisfying all conditions within a partitioned group combined with a sub-linear model that provide the greatest reduction in overall model error. If two or more conditions apply to a case, the final values are averaged. In this study, the *Cubist* statistical package (Kuhn et al. 2014) using software R (R Development Core Team 2015) was used to implement the RT algorithm. To act as training cases to guide the mapping of T_a , 30 covariate raster layers (GeoTIFFs) were produced. The majority of the raster layers were derived from the 3-arc sec (80-m resolution) Shuttle Radar Topography Mission (SRTM) DEM (Gallant et al. 2011) using the GIS software SAGA version 2.08 (Conrad et al. 2015) to form the following covariates: multi-resolution valley bottom flatness (MrVBF) (Gallant and Dowling 2003); multiresolution ridge-top flatness (MrRTF) (Gallant and Dowling 2003); topographic wetness index; SAGA wetness index (SWI); vertical distance to channel network; solar insolation indices (direct/diffuse/duration/total insolation); valley depth; slope; slope height; midslope position; sky view factor and visible sky indices; wind effect indexes (windward/leeward effect); wind speed/direction; analytical hill shade; northness/eastness index (as a surrogate for aspect); effective air flow heights; diurnal anisotropic heating; elevation above channel network; LS-factor; modified catchment area; longitude; latitude; distance to sea. A resampled version of a persistent greenness index was also produced separately from Landsat 8 satellite imagery and included as part of the covariate suite of raster layers. The raster covariate layers were then placed in a raster stack and spatially intersected with the geographic locations of the T_a predictions, namely the locations of the BoM weather stations and logger sites. The data presented in this manner made it possible to form Cubist models of T_a using the BoM observations along with the calibrated T_a estimates (generated from the calibration process) at the logger sites on an hourly basis. Thus, T_a

estimates along with the covariate values acted as training cases in the RT interpolation (refer to parts two and three in Fig. 2). To parameterise the Cubist model, the maximum allowable number of trees to grow was set to 100 and for unbiased rules to be formed. This was in addition to allowing the Cubist model to predict up to 100% outside of its training range. This was done to model T_a at the extreme ends of the covariate distribution range that would otherwise be unaccounted in the training cases. Furthermore, the algorithm was allowed to produce up to ten committee models as a way to ‘boost’ and generalise the models, which was capable of reducing error by up to 5%, as shown in Webb et al. (2018). To map modelled T_a predictions across the state, the predictions were spatially interpolated using the covariate stack along with the model parameters to guide the predictions at a spatial resolution of 80 m (equivalent to the SRTM DEM resolution). The result was 8760 T_a maps produced for every hour during the evaluation period from 23 September 2014 to 22 September 2015.

For TPS interpolation, its application involved a trivariate approach of incorporating latitude, longitude, and elevation, as per Jeffrey et al. (2001). The algorithm accounts for the temperature variation via a partial spline of the independent variables of latitude and longitude combined with a linear dependence on elevation to account for the temperature lapse rates (Hutchinson 1991). The smoothing parameter of the spline is parametrised via minimising the generalised cross validation (GCV), calculated by removing each data point and summing the squared error that transpires from the resulting surface. For application in this study, the *Fields* statistical package (Nychka et al. 2017) was used to implement the algorithm in R. As similarly performed with the Cubist procedure, the SRTM DEM were spatially intersected with the geographic locations (latitude and longitude) of the T_a predictions. This provided a consistent template to routinely form TPS models on an hourly basis, i.e. using the BoM observations along with the calibrated T_a estimates from the logger sites as the data points. To map modelled T_a predictions across the state, the predictions were spatially interpolated using the DEM as the z-covariate along with the inherent cell centroids providing the coordinate parameters of latitude and longitude required for the spline smoothing application to be applied continuously. Similar to the RT interpolation, 8760 T_a maps produced for every hour during the evaluation period from 23 September 2014 to 22 September 2015 at the equivalent spatial resolution of 80 m.

2.6 Evaluating the calibration and interpolation procedures

The performance of the calibration procedure was evaluated in the second year (23 September 2014 to 22 September 2015, i.e. the evaluation period) by cross-validating the calibrated

BoM grid predictions against actual T_a values at the logger sites (refer to part 2, Fig. 2). Validation metrics used to assess the accuracy of the calibration method included the mean absolute error (MAE), root-mean-square error (RMSE), coefficient of determination (R^2) and the concordance coefficient. The concordance coefficient (p_c) was used to assess agreement between model and actual data that fall on the 45° line through the origin, as defined by Lin (1989):

$$p_c = \frac{2p\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2} \quad (3)$$

where for μ_x and μ_y represent the means for x and y , respectively, σ_x^2 and σ_y^2 represent the corresponding variances, and p is the correlation coefficient between x and y . A concordance rating close to one indicates strong agreement between modelled/actual T_a pairings that fall on the 45° line through the origin.

To assess the performance of the RT and TPS interpolation, a leave-one-out cross-validation procedure was performed for each logger site. Specifically, the training dataset was split into i parts where i is equal to the number of loggers used in the analysis, i.e. 267. For each logger in i , the i th logger was kept for validation (i.e. using actual logger recordings from the evaluation period) while the remaining dataset ($i - 1$), comprising of BoM recordings and calibrated T_a predictions at the logger sites was used for RT and TPS modelling to predict T_a at the i th logger site. This was performed for each hour in the evaluation period to produce a set of modelled RT and TPS estimates versus actual logger recordings at each site. Validation metrics used to assess modelling accuracy included the MAE, RMSE, coefficient of determination (R^2) and the concordance coefficient.

In addition, both the calibration and interpolation validation metrics were stratified by landform type using the topographic position index (TPI) developed by Guisan et al. (1999), Jenness (2006) and Weiss (2001). This was done to assess prediction accuracy under different topographic settings, which was likely to vary owing to the influence terrain has on temperature lapse rates, particularly that of high complexity (Rolland 2003; Stahl et al. 2006). The algorithm works by comparing the elevation of each cell in a DEM to the mean elevation of a specified neighbourhood of adjacent cells to produce a TPI value. Positive TPI values represent locations that are higher than the average of their surroundings (e.g. hills and ridges), whereas negative TPI values represent locations that are lower (e.g. valleys and canyons). Near zero values represent flat areas such as plains. To form the TPI, the SRTM DEM was used in the landform classifications module using GIS software SAGA version 2.08 (Conrad et al. 2015). The default settings were chosen that included a neighbourhood search radius of 100 m with no distance weighting function applied. The result was a raster layer comprising of ten landform classifications (Fig. 3). These included

canyons/deeply incised streams (0), midslope drainages/shallow valleys (3), upland drainages/headwaters (0), U-shaped valleys (9), plains (179), open slopes (58), upper slopes/mesas (9), local ridges/hills in valleys (0), midslope ridges/small hills in plains (5), and mountain tops/high ridges (4). Note that the values in brackets denote the number of logger sites found in each classification.

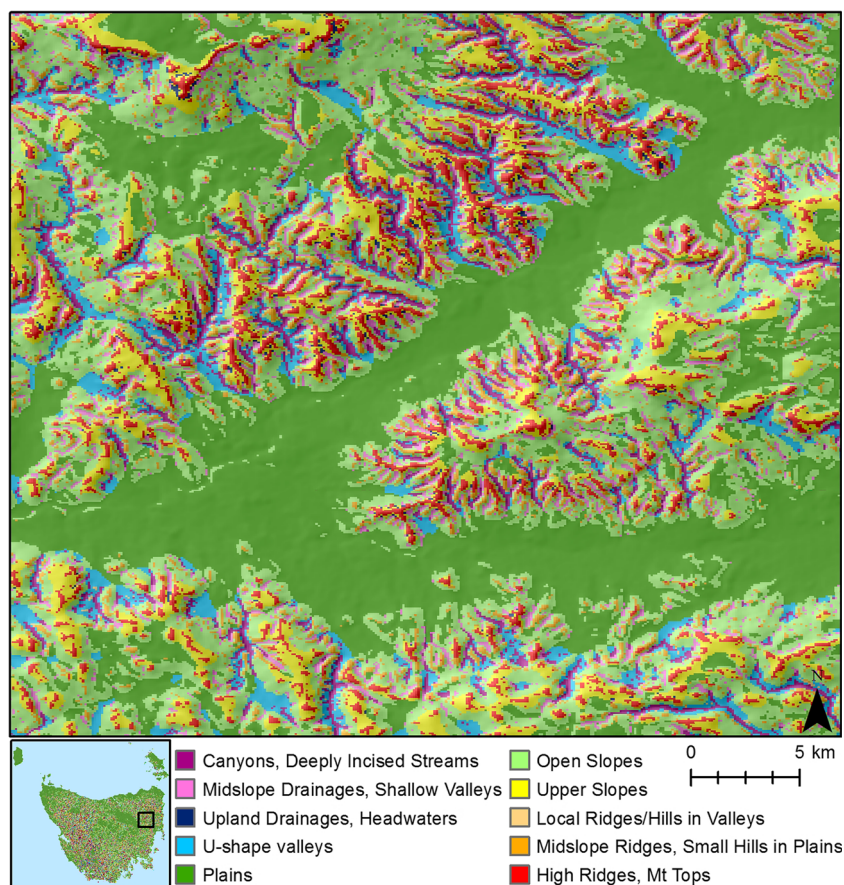
2.7 Application to real-time monitoring of T_a

To assess the practicality of applying the proposed methodology to operational real-time monitoring of hourly T_a , the method was automated using software R (R Development Core Team 2015) to create a fully automated T_a mapping system. The system was trialled over a 7-day period from 7 to 13 February 2020, using live BoM observations to drive the system. The program consisted of four components, firstly the import of live BoM data via the Internet; secondly, the formation of BoM grids based on the live data; thirdly, the calibration of the BoM grid estimates using the calibration equations derived from the calibration period (refer 'section 2.4'); and finally, the mapping of the calibrated BoM grid predictions (along with the BoM data observations) using RT and TPS interpolation (refer 'section 2.5'). The entire process is depicted in part 3 of Fig. 2.

Live BoM data were automatically downloaded for individual station observations every hour from the BoM observations portal (e.g. <http://www.bom.gov.au/fwo/IDT60801/IDT60801.<stationIDnumber>.axf>). These data were accessed as a comma delimited text file which were routinely updated every hour with an approximate time lag ranging from 5 to 25 min from when the observation was recorded (the timeliness of the updates varied from station to station). The mapping system was programmed to query and import recordings relevant to the nearest hour. However, because of the observational time lags, the system was programmed to make queries at 5, 10, 15, 20 and 29 min past the hour of interest. A threshold was set where at least 41 out of the 43 BoM stations that were used in the calibration and evaluation analysis have available observations before the mapping was allowed to commence. This was to instill integrity into the system and to limit the number of missing observations when forming the BoM grids that could otherwise produce inaccuracies into the final mapped output. A cut-off time limit was set at 29 min past the hour to abort the mapping if the threshold of 41 observations was not met (resulting in mapping not being conducted for that hour).

To map RT and TPS predictions across the state, the Raster package (Hijmans and van Etten 2012) was used to map the hourly predictions in a continuous manner in combination to the Cubist (Kuhn et al. 2014) and TPS (Nychka et al. 2017) algorithms. The clusterR function within the Raster package was then used to improve the processing speed of both

Fig. 3 Mapped landform classification categories for Tasmania (example magnified to the extent of the Fingal Valley region)



algorithms in R, which enabled mapping to occur using multi-core processors in parallel. In this manner, outputs were processed sequentially, first using Cubist then TPS thereafter to produce separate maps corresponding to each interpolation method. The entire mapping system was hosted on a cloud-based computing instance provided by the Australian National eResearch Collaboration Tools and Resources project (NeCTAR) with node specifications comprising 16 virtual CPU cores and 64GB RAM. Spatial outputs were saved as individual GeoTIFF raster files at a grid cell resolution of 80 m, i.e. equivalent to the SRTM DEM resolution.

3 Results

3.1 Assessment of the calibration procedure

To assess the error of the calibration procedure, MAE and RMSE were calculated across the seasons in the evaluation period (Sept 2014–Sept 2015) and listed in Table 1. The error was broadly similar across the seasons. MAE values ranged between 1.02 °C in summer and 0.98 °C in spring, and RMSE values ranged between 1.33 °C in summer and 1.29 °C in winter (Table 1). This was in addition to having high R^2 and P_c with values above 0.8 indicating that the modelled/actual

data pairings were strongly correlated and highly associated with the 45° line through the origin. This also suggested that predictions were relatively stable across the evaluation period and did not vary greatly on a seasonal basis. Overall, winter and spring were slightly better suited to the calibration procedure, particularly when compared to summer, registering an RMSE improvement of 0.04 °C (i.e. 1.33–1.29 °C) and 0.03 °C (i.e. 1.33–1.3 °C) for winter and spring, respectively. This was despite the R^2 values for summer (0.92) being superior compared to the R^2 and for winter (0.88).

When the calibration accuracy was assessed against landform type, predictions were prone to greater inaccuracies for loggers located on mountaintops/high ridges (MT) and upper slope/mesa areas (US). These registered MAEs of 1.33 °C, 1.1 °C, 1.07 °C and 1.18 °C for MT; and 1.25 °C, 1.11 °C, 1.12 °C and 1.18 °C for US across summer, autumn, winter and spring, respectively (Table 2). Out of these results, it was clear that predictions for summer tended to be more inaccurate compared to other seasons. Taking into account the remaining landform types of midslope ridges/small hills in plains (MR), open slopes (OS), plains (PL), midslope drainages/shallow valleys (SV) and U-shaped valleys (UV), these were more prone to greater discrepancies in summer compared to other seasons. PL had the most accurate predictions on average across all seasons with MAE's below 1 °C (i.e. 0.98 °C,

Table 1 Validation statistics for the logger calibration procedure showing the coefficient of determination (R^2), concordance coefficient (P_c), mean absolute error (MAE, °C) and root-mean-square error (RMSE, °C) averaged for each logger site according to the season

	Summer	Autumn	Winter	Spring
R^2				
mean	0.92	0.9	0.88	0.91
min	0.72	0.68	0.53	0.73
max	0.98	0.97	0.97	0.97
SD	0.04	0.04	0.06	0.04
P_c				
mean	0.96	0.95	0.96	0.95
min	0.84	0.82	0.84	0.84
max	0.99	0.99	0.99	0.99
SD	0.02	0.02	0.02	0.02
MAE				
mean	1.02	1.01	0.99	0.98
min	0.46	0.46	0.49	0.52
max	2.47	2.26	2.2	2.11
sd	0.24	0.24	0.25	0.22
RMSE				
mean	1.33	1.33	1.29	1.3
min	0.64	0.63	0.64	0.7
max	3.06	2.89	2.76	2.69
sd	0.3	0.32	0.31	0.28

sd, standard deviation; min, minimum; max, maximum

0.98 °C, 0.96 °C and 0.95 °C for summer, autumn, winter and spring, respectively. U-shaped valleys (UV) also performed well and exhibited MAE's of 0.97 °C for autumn, winter and spring. Notably, loggers located in shallow valleys (SV) during winter had the lowest MAE and RMSE values of 0.91 °C and 1.19 °C, respectively. Nearly all R^2 and P_c values were above 0.8 indicating strong correlation and concordance with the validation data regardless of landform type, except R^2 for US in winter registering a value of 0.79.

When viewed over a 24-h period across the seasons (Fig. 4), the calibration procedure was prone to greater inaccuracies during summer when compared to the other seasons, particularly during the night. This was especially true for MT for hours between 11 pm and 6 am, which consistently registered MAE's above 1.4 °C. A clear downward trend was exhibited for MAE concerning hours in the late afternoon and early evening, typically between 3 pm and 8 pm where predictions gradually became more accurate heading into the evening. This trend was notable across all seasons. Of particular note was for US in summer, where MAE generally peaked at 1 pm (1.51 °C) and steadily decreased into the early evening with the minimum MAE reached at 8 pm (0.93 °C) before steadily rising again into the late evening. To a lesser extent, this was also evident for hours between 6 am to 9 am. For example, for

MT in summer, the MAE peaked at 6 am (1.57 °C) and decreased to 1.23 °C by 10 am before steadily rising again in the late morning and early afternoon.

3.2 Assessment of the RT and TPS interpolation

Hourly T_a data from BoM sites along with the calibrated estimates produced for the logger sites were interpolated across Tasmania using RT and TPS interpolation. For predictions generated from the RT interpolation procedure, the MAE and RMSE values were only slightly worse than the calibration process (Table 1) with MAE difference of 0.04 °C, 0.04 °C, 0.03 °C and 0.05 °C, and RMSE of 0.06 °C, 0.06 °C, 0.05 °C and 0.03 °C for summer, autumn, winter and spring, respectively (Table 3). A similar trend was encountered by the TPS interpolation with MAE difference of 0.04 °C, 0.06 °C and 0.02 °C, and RMSE of 0.09 °C, 0.11 °C and 0.4 °C for autumn, winter and spring, respectively. For summer, however, the MAE improved by 0.01 °C, although the RMSE deteriorated by 0.02 °C. With regard to the R^2 and P_c values, both interpolation methods had differences, deteriorating by no more than 0.01 °C when compared to the R^2 and P_c values from the calibration process. On the whole, this indicated that both interpolation methods did not significantly degrade the calibrated estimates when used to map T_a across the state.

When comparing between RT and TPS (Table 3), both methods yielded similar results with MAE's of 1.06 °C, 1.05 °C, 1.02 °C and 1.03 °C for the RT method versus 1.01 °C, 1.05 °C, 1.05 °C and 1 °C for the TPS method across summer, autumn, winter and spring, respectively. Similarly, the RMSE was 1.39 °C, 1.39 °C, 1.34 °C and 1.36 °C for the RT method versus 1.35 °C, 1.42 °C, 1.4 °C and 1.34 °C for the TPS method. A closer look, however, reveals a slight improvement for TPS interpolation over RT in summer and spring with the MAE improving by 0.05 °C and 0.03 °C (RMSE difference of 0.04 °C and 0.02 °C), respectively. Conversely, RT was slightly better in predicting winter temperatures compared to TPS with the MAE and RMSE improving by 0.03 °C and 0.06 °C, respectively. The MAE for autumn both registered the same MAE of 1.05 °C, although the RMSE was slightly improved for RT by 0.03 °C.

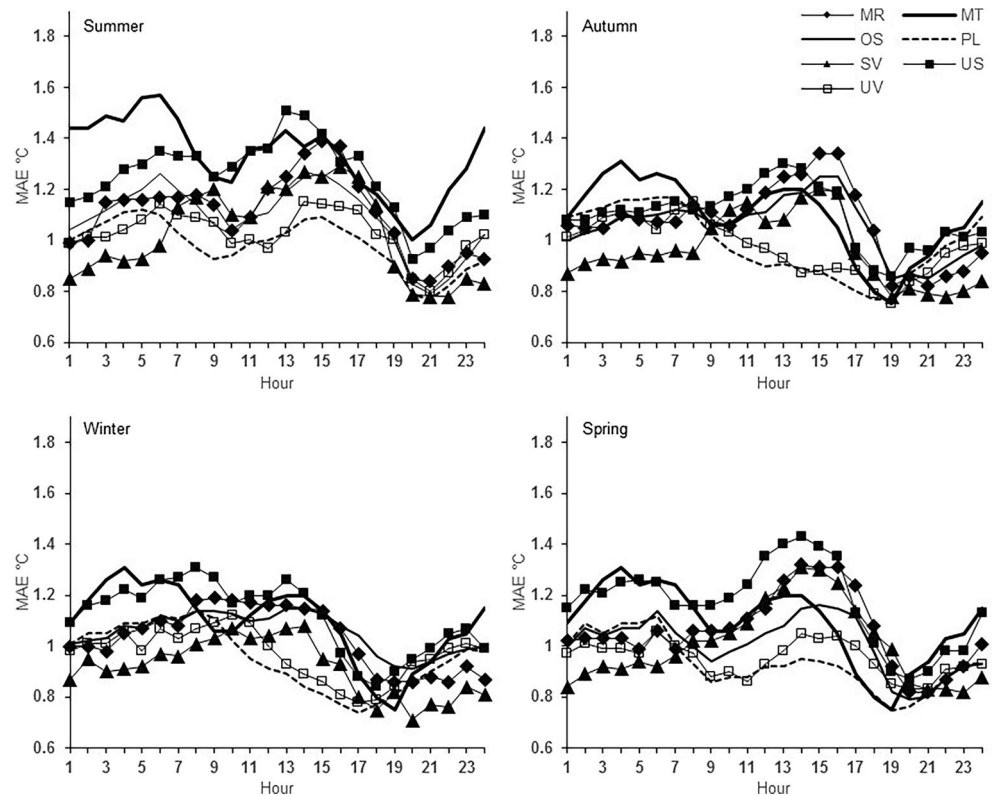
In regards to interpolation versus landform type, it was clear that the RT method (Table 4) was less effective in predicting T_a during summer compared to TPS (Table 5). The MAE for TPS improved by 0.13 °C, 0.24 °C, 0.08 °C, 0.01 °C, 0.07 °C, 0.13 °C and 0.11 °C for landform types concerning MR, MT, OS, PL, SV, US and UV, respectively, when compared to the corresponding MAE's for RT. The same was encountered for spring with the corresponding MAE for TPS improving by 0.07 °C, 0.06 °C, 0.06 °C, 0.01 °C, 0.07 °C, 0.06 °C and 0.4 °C, respectively. Conversely, RT was more accurate than TPS in winter with MAE

Table 2 Validation statistics for the logger calibration procedure showing the coefficient of determination (R^2), concordance coefficient (P_c), mean absolute error (MAE, °C) and root-mean-square error (RMSE, °C) averaged for each logger site according to season and stratified by landform type

	Summer				Autumn				Winter				Spring			
	mean		sd		mean		sd		mean		sd		mean		sd	
	min	max	min	max	min	max	min	max	min	max	min	max	min	max	min	max
R^2																
MR	0.92	0.94	0.01	0.01	0.88	0.87	0.01	0.01	0.84	0.82	0.01	0.01	0.9	0.88	0.03	0.02
MT	0.9	0.93	0.04	0.04	0.86	0.82	0.03	0.03	0.8	0.77	0.03	0.03	0.89	0.85	0.05	0.03
OS	0.91	0.96	0.05	0.05	0.89	0.74	0.04	0.04	0.87	0.68	0.04	0.04	0.9	0.74	0.05	0.04
PL	0.93	0.98	0.03	0.03	0.91	0.7	0.04	0.04	0.89	0.53	0.04	0.04	0.92	0.78	0.05	0.03
SV	0.94	0.94	0	0	0.89	0.87	0.03	0.03	0.86	0.85	0.03	0.03	0.92	0.9	0.01	0.01
US	0.9	0.95	0.06	0.06	0.86	0.68	0.07	0.07	0.79	0.63	0.07	0.07	0.88	0.73	0.08	0.06
UV	0.93	0.96	0.02	0.02	0.91	0.86	0.03	0.03	0.88	0.83	0.03	0.03	0.92	0.9	0.04	0.02
P_c																
MR	0.96	0.97	0.01	0.01	0.93	0.93	0.01	0.01	0.91	0.89	0.01	0.01	0.95	0.93	0.01	0.01
MT	0.94	0.96	0.03	0.03	0.92	0.9	0.02	0.02	0.88	0.86	0.02	0.02	0.94	0.92	0.03	0.01
OS	0.95	0.98	0.03	0.03	0.94	0.84	0.03	0.03	0.92	0.8	0.03	0.03	0.95	0.84	0.03	0.03
PL	0.96	0.99	0.02	0.02	0.95	0.83	0.02	0.02	0.94	0.68	0.02	0.02	0.96	0.87	0.03	0.02
SV	0.97	0.97	0	0	0.94	0.93	0.01	0.01	0.93	0.92	0.01	0.01	0.95	0.95	0.01	0.01
US	0.94	0.97	0.04	0.04	0.92	0.82	0.04	0.04	0.87	0.74	0.04	0.04	0.93	0.85	0.06	0.04
UV	0.96	0.98	0.01	0.01	0.95	0.93	0.02	0.02	0.93	0.88	0.02	0.02	0.96	0.94	0.03	0.01
MAE																
MR	1.11	1.26	0.13	0.13	1.08	1.01	0.06	0.06	1.04	0.95	0.06	0.06	1.08	0.92	0.09	0.1
MT	1.33	1.81	0.44	0.44	1.1	0.74	0.28	0.28	1.07	0.7	0.28	0.28	1.18	0.84	0.28	0.31
OS	1.08	2.47	0.27	0.27	1.06	0.54	0.25	0.25	1.05	0.49	0.25	0.25	1.03	0.57	0.27	0.22
PL	0.98	1.89	0.21	0.21	0.98	0.46	0.25	0.25	0.96	0.49	0.25	0.25	0.95	0.52	0.25	0.21
SV	1.04	1.18	0.12	0.12	0.97	0.86	0.11	0.11	0.91	0.79	0.11	0.11	1.02	0.91	0.18	0.12
US	1.25	1.51	0.22	0.22	1.11	0.83	0.21	0.21	1.12	0.81	0.21	0.21	1.18	0.83	0.19	0.24
UV	1.02	1.41	0.18	0.18	0.97	0.71	0.17	0.17	0.97	0.69	0.17	0.17	0.97	0.76	0.19	0.14
RMSE																
MR	1.43	1.61	0.16	0.16	1.4	1.35	0.04	0.04	1.37	1.27	0.04	0.04	1.41	1.23	0.11	0.11
MT	1.65	2.22	0.52	0.52	1.43	0.96	0.34	0.34	1.38	0.89	0.34	0.34	1.51	1.08	0.36	0.37
OS	1.41	3.06	0.33	0.33	1.36	0.71	0.31	0.31	1.35	0.69	0.31	0.31	1.35	0.79	0.31	0.28
PL	1.29	2.51	0.28	0.28	1.31	0.63	0.33	0.33	1.27	0.64	0.33	0.33	1.26	0.7	0.32	0.28
SV	1.36	1.57	0.18	0.18	1.28	1.18	0.14	0.14	1.19	1.06	0.14	0.14	1.23	1.23	0.23	0.13
US	1.58	1.87	0.25	0.25	1.44	1.08	0.28	0.28	1.44	0.98	0.28	0.28	1.52	1.07	0.25	0.29
UV	1.32	1.85	0.24	0.24	1.28	0.92	0.23	0.23	1.28	0.9	0.23	0.23	1.28	1	0.23	0.18

MR, midslope ridges/small hills in plains; MT, mountain tops/high ridges; OS, open slopes; PL, plains; SV, midslope drainages/shallow valleys; US, upper slopes/mesas; UV, U-shaped valleys; sd, standard deviation; min, minimum; max, maximum

Fig. 4 Mean absolute error (MAE, °C) over a 24-h period for the calibration procedure averaged for each logger site according to season and landform type: MR, midslope ridges/small hills in plains; MT, mountain tops/high ridges; OS, open slopes, PL, plains; SV, midslope drainages/shallow valleys; US, upper slopes/mesas; UV, U-shaped valleys



improvements of 0.03 °C, 0.27 °C, 0.02 °C, 0.03 °C, 0.01 °C and 0.01 °C for MR, MT, OS, PL, US and UV, respectively.

Table 3 Validation statistics for the RT and TPS interpolation procedure showing the coefficient of determination (R^2), concordance coefficient (P_c), mean absolute error (MAE, °C) and root-mean-square error (RMSE, °C) averaged for each logger site according to the season

	Summer		Autumn		Winter		Spring	
	RT	TPS	RT	TPS	RT	TPS	RT	TPS
R^2								
mean	0.92	0.93	0.9	0.9	0.87	0.88	0.91	0.92
min	0.74	0.72	0.67	0.58	0.5	0.35	0.71	0.72
max	0.97	0.98	0.97	0.99	0.96	0.98	0.97	0.98
sd	0.04	0.04	0.04	0.05	0.06	0.06	0.03	0.04
P_c								
mean	0.95	0.96	0.94	0.94	0.95	0.96	0.95	0.95
min	0.81	0.83	0.82	0.76	0.81	0.83	0.83	0.84
max	0.98	0.99	0.98	0.99	0.98	0.99	0.98	0.99
sd	0.02	0.03	0.03	0.04	0.02	0.03	0.02	0.03
MAE								
mean	1.06	1.01	1.05	1.05	1.02	1.05	1.03	1
min	0.62	0.38	0.56	0.39	0.51	0.37	0.55	0.4
max	2.49	2.61	2.27	2.36	2.47	2.56	2.09	2.15
sd	0.24	0.26	0.24	0.3	0.26	0.32	0.24	0.26
RMSE								
mean	1.39	1.35	1.39	1.42	1.34	1.4	1.36	1.34
min	0.8	0.53	0.72	0.51	0.65	0.5	0.72	0.55
max	3.08	3.24	2.87	3.04	2.9	3.02	2.6	2.7
sd	0.3	0.34	0.33	0.42	0.33	0.42	0.3	0.36

sd, standard deviation; min, minimum; max, maximum

In addition, the landforms concerning MT, PL, SV, US and UV performed better using RT than TPS in autumn with MAE improving by 0.13 °C, 0.01 °C, 0.01 °C, 0.02 °C and 0.01 °C, respectively. However, MR and OS improved by 0.02 °C and 0.01 °C, respectively when using the TPS over the RT method in autumn. It should be noted that TPS predictions made in summer for landform types concerning MR, MT, OS and US, the error actually improved by 0.09 °C, 0.24 °C, 0.07 °C and 0.11 °C, respectively for MAE and 0.09 °C, 0.24 °C, 0.07 °C and 0.09 °C, respectively for RMSE when compared to the calibrated estimates (Table 2). This was also encountered for OS in autumn with MAE improving by 0.01 °C, in addition to MR, OS and US in spring with MAE improving by 0.07 °C, 0.04 °C and 0.02 °C, respectively.

In regards to RT interpolation versus time of day, it was clear that the RT algorithm was more prone to larger MAE's for MT areas, and to a lesser extent US areas, during the middle part of the day in summer and spring. This was particularly apparent between 12 pm and 3 pm with peak error at 1 pm in summer registering 1.83 °C (Fig. 5). Comparatively, the TPS was more adept at predicting T_a during summer in MT areas where the MAE reduced by as much as 0.54 °C during the middle part of the day at 1 pm (i.e. 1.83 °C versus 1.29 °C) (Fig. 6). However, during early morning between midnight and 7 am in autumn, winter and spring, it was clear that the TPS method was less accurate in MT areas, with the MAE peaking at 1.91 °C at 8 am in winter, compared to RT where MAE recorded 1.3 °C, i.e. an MAE difference of 0.61

Table 4 Validation statistics for the RT interpolation procedure showing the coefficient of determination (R^2), concordance coefficient (P_c), mean absolute error (MAE, °C) and root-mean-square error (RMSE, °C) averaged for each logger site according to season and stratified by landform type.

	Summer				Autumn				Winter				Spring			
	mean	min	max	sd	mean	min	max	sd	mean	min	max	sd	mean	min	max	sd
R^2																
MR	0.92	0.9	0.93	0.01	0.89	0.87	0.91	0.01	0.85	0.81	0.89	0.03	0.91	0.88	0.94	0.02
MT	0.89	0.85	0.93	0.03	0.87	0.82	0.91	0.04	0.81	0.77	0.85	0.04	0.89	0.85	0.91	0.02
OS	0.91	0.74	0.95	0.04	0.9	0.75	0.95	0.04	0.87	0.68	0.95	0.05	0.9	0.74	0.95	0.04
PL	0.92	0.8	0.97	0.03	0.9	0.67	0.97	0.04	0.88	0.62	0.96	0.05	0.92	0.77	0.97	0.03
SV	0.93	0.92	0.93	0.01	0.88	0.86	0.91	0.03	0.86	0.85	0.87	0.01	0.91	0.9	0.91	0.01
US	0.9	0.74	0.95	0.06	0.86	0.68	0.91	0.07	0.77	0.5	0.89	0.11	0.88	0.71	0.94	0.07
UV	0.92	0.89	0.95	0.02	0.9	0.86	0.94	0.03	0.87	0.81	0.93	0.04	0.91	0.89	0.94	0.02
P_c																
MR	0.95	0.94	0.97	0.01	0.93	0.92	0.94	0.01	0.9	0.89	0.91	0.01	0.94	0.93	0.96	0.01
MT	0.93	0.91	0.96	0.02	0.91	0.89	0.94	0.02	0.87	0.84	0.9	0.03	0.92	0.89	0.95	0.03
OS	0.95	0.81	0.98	0.03	0.94	0.84	0.97	0.02	0.92	0.79	0.96	0.03	0.94	0.85	0.97	0.02
PL	0.96	0.88	0.98	0.02	0.94	0.82	0.98	0.03	0.93	0.79	0.98	0.03	0.95	0.86	0.98	0.02
SV	0.96	0.94	0.96	0.01	0.93	0.91	0.95	0.02	0.92	0.9	0.93	0.01	0.94	0.93	0.95	0.01
US	0.94	0.84	0.97	0.04	0.92	0.82	0.95	0.04	0.86	0.69	0.94	0.07	0.93	0.83	0.96	0.04
UV	0.95	0.92	0.98	0.02	0.94	0.91	0.96	0.02	0.91	0.86	0.95	0.03	0.95	0.93	0.97	0.01
MAE																
MR	1.15	0.93	1.39	0.18	1.11	1.01	1.2	0.08	1.12	0.99	1.18	0.08	1.08	0.89	1.17	0.11
MT	1.33	0.85	1.59	0.33	1.12	0.84	1.37	0.22	1.13	1.01	1.35	0.15	1.27	0.84	1.59	0.34
OS	1.08	0.7	2.49	0.27	1.06	0.67	2.27	0.23	1.03	0.72	2.18	0.22	1.05	0.64	2.09	0.23
PL	1.02	0.62	2.18	0.21	1.03	0.56	2.02	0.25	1	0.51	2.47	0.27	1	0.55	2.01	0.23
SV	1.19	1.01	1.47	0.25	1	0.86	1.16	0.15	0.92	0.81	1.07	0.14	1.12	0.93	1.33	0.2
US	1.27	1.03	1.53	0.2	1.14	0.74	1.43	0.2	1.18	0.75	1.51	0.22	1.22	0.93	1.56	0.23
UV	1.17	0.86	1.74	0.28	1.14	0.91	1.64	0.22	1.2	0.85	1.78	0.25	1.07	0.86	1.43	0.17
RMSE																
MR	1.49	1.23	1.8	0.21	1.45	1.32	1.53	0.1	1.44	1.34	1.55	0.08	1.42	1.18	1.52	0.14
MT	1.71	1.08	2.04	0.43	1.48	1.12	1.8	0.28	1.47	1.33	1.76	0.2	1.65	1.11	2.06	0.43
OS	1.42	0.93	3.08	0.34	1.38	0.88	2.87	0.3	1.33	0.93	2.77	0.29	1.38	0.86	2.6	0.29
PL	1.35	0.8	2.55	0.28	1.38	0.72	2.68	0.35	1.32	0.65	2.9	0.35	1.33	0.72	2.36	0.31
SV	1.53	1.33	1.89	0.31	1.35	1.18	1.62	0.24	1.23	1.06	1.48	0.22	1.49	1.24	1.8	0.28
US	1.6	1.33	1.93	0.23	1.47	1.01	1.81	0.24	1.53	1	1.97	0.28	1.56	1.19	2	0.29
UV	1.53	1.13	2.21	0.36	1.49	1.19	2.03	0.26	1.56	1.17	2.32	0.32	1.42	1.13	1.82	0.21

MR, midslope ridges/small hills in plains; MT, mountain tops/high ridges; OS, open slopes; PL, plains; SV, midslope drainages/shallow valleys; US, upper slopes/mesas; UV, U-shaped valleys; sd, standard deviation; min, minimum; max, maximum

Table 5 Validation statistics for the TPS interpolation procedure showing the coefficient of determination (R^2), concordance coefficient (P_c), mean absolute error (MAE, °C) and root-mean-square error (RMSE, °C) averaged for each logger site according to season and stratified by landform type

	Summer				Autumn				Winter				Spring			
	mean	min	max	sd	mean	min	max	sd	mean	min	max	sd	mean	min	max	sd
R^2																
MR	0.94	0.92	0.96	0.01	0.91	0.87	0.93	0.02	0.87	0.84	0.9	0.02	0.93	0.92	0.94	0.01
MT	0.93	0.9	0.96	0.03	0.89	0.86	0.91	0.02	0.83	0.81	0.85	0.02	0.9	0.89	0.92	0.01
OS	0.93	0.72	0.96	0.05	0.91	0.72	0.96	0.04	0.89	0.65	0.96	0.05	0.92	0.72	0.96	0.04
PL	0.93	0.78	0.98	0.04	0.9	0.58	0.99	0.05	0.88	0.35	0.98	0.06	0.92	0.75	0.98	0.04
SV	0.93	0.92	0.94	0.01	0.87	0.83	0.92	0.04	0.85	0.83	0.89	0.03	0.91	0.91	0.91	0
US	0.91	0.8	0.95	0.05	0.87	0.72	0.92	0.06	0.8	0.66	0.9	0.07	0.89	0.76	0.95	0.06
UV	0.93	0.87	0.96	0.03	0.9	0.85	0.93	0.03	0.87	0.82	0.91	0.03	0.92	0.86	0.94	0.03
P_c																
MR	0.96	0.95	0.97	0.01	0.93	0.92	0.95	0.01	0.9	0.87	0.94	0.03	0.96	0.95	0.96	0
MT	0.96	0.94	0.96	0.01	0.9	0.8	0.94	0.07	0.84	0.68	0.91	0.11	0.93	0.91	0.95	0.02
OS	0.96	0.83	0.98	0.03	0.94	0.81	0.98	0.03	0.92	0.75	0.97	0.04	0.95	0.84	0.98	0.03
PL	0.96	0.87	0.99	0.03	0.94	0.76	0.99	0.04	0.92	0.59	0.99	0.05	0.95	0.84	0.99	0.03
SV	0.96	0.95	0.97	0.01	0.93	0.91	0.96	0.03	0.92	0.91	0.94	0.02	0.95	0.94	0.95	0.01
US	0.95	0.89	0.97	0.03	0.92	0.84	0.94	0.03	0.87	0.77	0.94	0.05	0.94	0.87	0.97	0.03
UV	0.96	0.92	0.98	0.02	0.93	0.89	0.96	0.02	0.9	0.83	0.95	0.04	0.95	0.92	0.97	0.02
MAE																
MR	1.02	0.86	1.34	0.2	1.09	0.96	1.17	0.09	1.15	0.92	1.44	0.21	1.01	0.99	1.06	0.03
MT	1.09	0.88	1.3	0.17	1.25	0.88	1.9	0.45	1.4	0.96	2.23	0.57	1.21	1.02	1.31	0.13
OS	1.01	0.59	2.61	0.31	1.05	0.53	2.36	0.28	1.05	0.53	2.25	0.28	0.99	0.55	2.15	0.28
PL	1	0.38	2.34	0.26	1.04	0.39	2.16	0.31	1.03	0.37	2.56	0.33	0.99	0.4	1.98	0.27
SV	1.12	0.98	1.37	0.22	1.01	0.89	1.17	0.15	0.92	0.81	1.08	0.14	1.05	0.95	1.24	0.16
US	1.14	0.9	1.41	0.21	1.16	0.84	1.43	0.21	1.19	0.72	1.83	0.32	1.16	0.82	1.43	0.21
UV	1.06	0.79	1.41	0.2	1.15	0.88	1.38	0.15	1.21	0.94	1.7	0.24	1.03	0.87	1.23	0.13
RMSE																
MR	1.34	1.13	1.73	0.23	1.45	1.24	1.62	0.14	1.52	1.2	1.8	0.27	1.34	1.27	1.46	0.09
MT	1.41	1.14	1.68	0.22	1.67	1.19	2.54	0.6	1.91	1.28	2.99	0.76	1.62	1.37	1.72	0.17
OS	1.34	0.88	3.24	0.39	1.38	0.72	3.04	0.38	1.37	0.71	2.94	0.36	1.32	0.81	2.7	0.36
PL	1.34	0.53	2.72	0.34	1.42	0.51	3	0.45	1.38	0.5	3.02	0.43	1.33	0.55	2.51	0.37
SV	1.45	1.24	1.8	0.31	1.37	1.16	1.62	0.24	1.22	1.05	1.45	0.21	1.39	1.23	1.7	0.26
US	1.49	1.2	1.78	0.23	1.56	1.12	1.95	0.3	1.6	0.95	2.47	0.44	1.53	1.04	1.86	0.27
UV	1.4	1.07	1.84	0.25	1.53	1.17	1.8	0.18	1.6	1.27	2.31	0.32	1.38	1.19	1.65	0.17

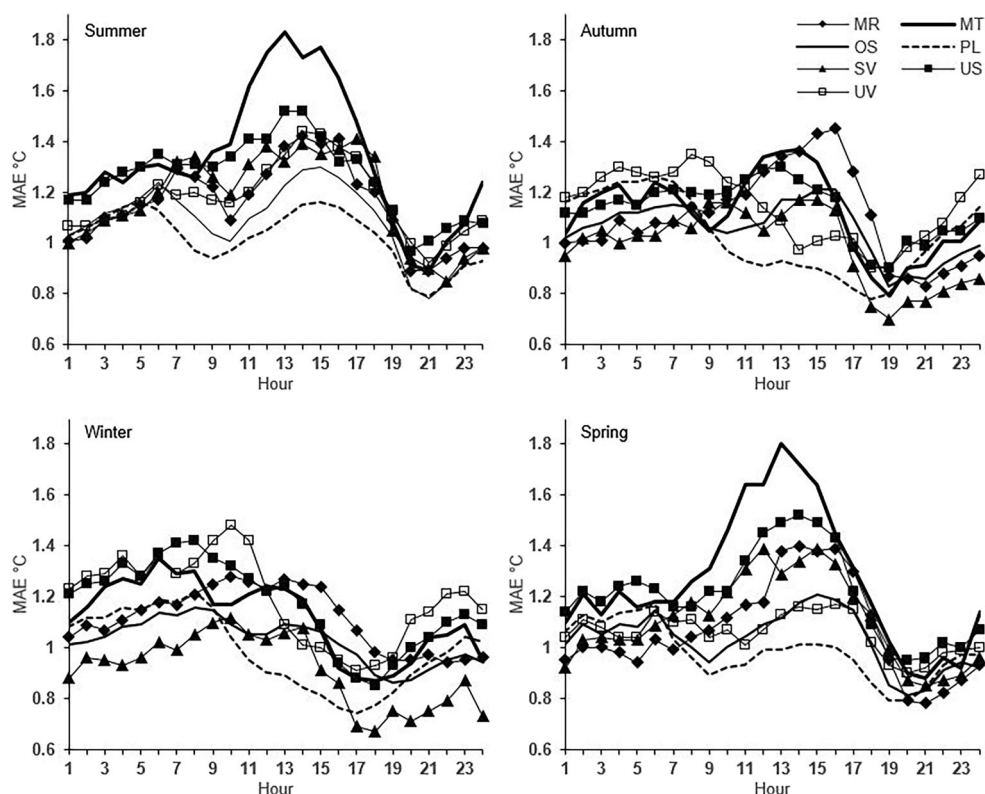
MR, midslope ridges/small hills in plains; MT, mountain tops/high ridges; OS, open slopes; PL, plains; SV, midslope drainages/shallow valleys; US, upper slopes/mesas; UV, U-shaped valleys; sd, standard deviation; min, minimum; max, maximum

°C. The TPS predictions tended to fluctuate wildly in winter decreasing abruptly from 8 am onwards with the MAE decreasing to a low 0.73 °C at 4 pm, compared to 0.92 °C for RT. For both methods, it was generally found that a downward trend concerning decreasing MAE for hours in the late afternoon and early evenings (as encountered for the calibrated BoM grid data, Fig. 4) was systemic across all seasons regardless of interpolation procedure. Similarly, an upward trend for steadily increasing MAE values also occurred in the late evenings and was common across all landforms and seasons.

3.3 Assessment of mapping T_a in near real time

The methodology was applied to mapping T_a in real time (refer to ‘section 2.7’) to produce hourly T_a maps based on RT and TPS interpolation over a 7-day period from 7 to 13 February 2020 (Fig. 7). In total, 168 hourly maps were produced using RT and TPS methods (Table 6). This was in line with the total number of hours within the 7-day period confirming that all possible hours were successfully processed by both interpolation methods. It also confirmed that the minimum number of allowable BoM stations (41)—a requirement

Fig. 5 Mean absolute error (MAE, °C) over a 24-h period for the RT interpolation procedure averaged for each logger site according to season and landform type: MR, midslope ridges/small hills in plains; MT, mountain tops/high ridges; OS, open slopes; PL, plains; SV, midslope drainages/shallow valleys; US, upper slopes/mesas; UV, U-shaped valleys



for the mapping to progress—was also met. Out of the hours processed, 39 maps were based on the minimum amount of BoM stations, 12 maps based on 42 stations and 117 based

on the maximum of 43 stations. In terms of the time taken to produce the RT and TPS maps (relevant to the start of the nearest recording hour), the majority (153) were successfully

Fig. 6 Mean absolute error (MAE, °C) over a 24-h period for the TPS interpolation procedure averaged for each logger site according to season and landform type: MR, midslope ridges/small hills in plains; MT, mountain tops/high ridges; OS, open slopes; PL, plains; SV, midslope drainages/shallow valleys; US, upper slopes/mesas; UV, U-shaped valleys

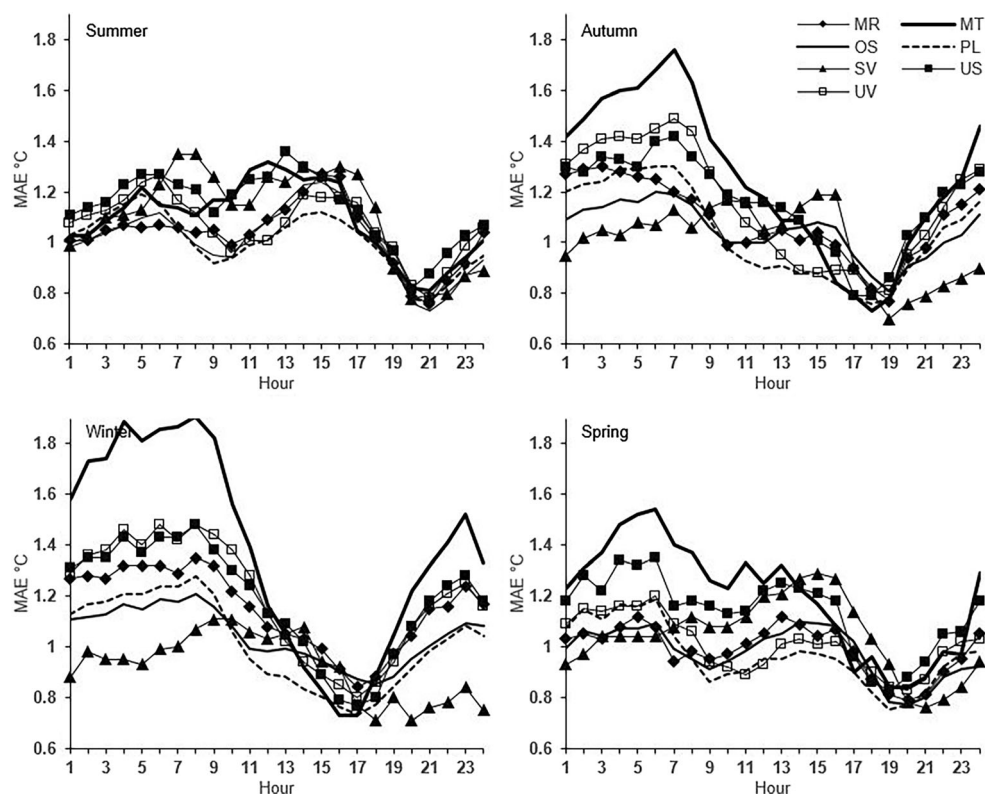
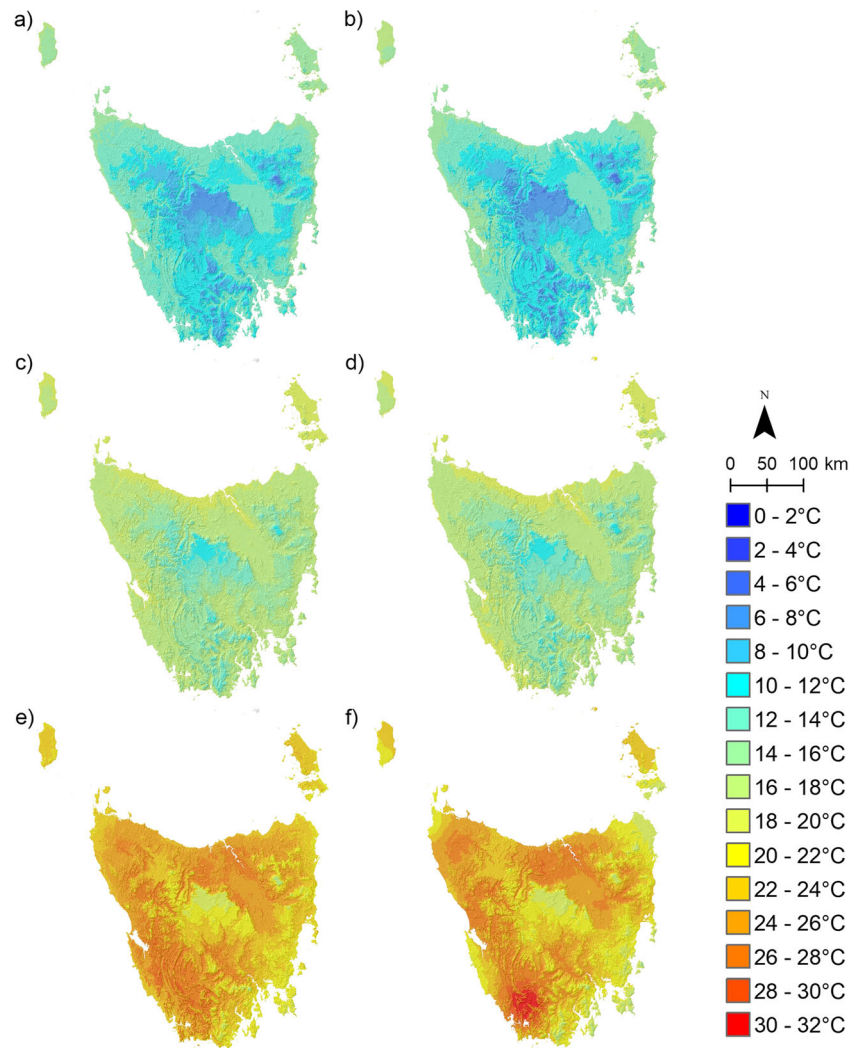


Fig. 7 T_a maps for selected hours showing the coldest (6 am, 9/02/2020 AEDT) for **a** RT- and **b** TPS-based maps, median (7 am, 11/02/2020 AEDT) for **c** RT- and **d** TPS-based maps, and warmest (4 pm, 11/02/2020 AEDT) for **e** RT- and **f** TPS-based maps in the 7-day period from 7 to 13 February 2020.



processed at the 28- and 17-min mark, respectively. For the BoM import time frequencies, most were completed at the 15-min mark (achieved 135 times) and were therefore highly correlated to processing time. Thus, a large portion of RT maps completed at the 27–28-min mark and TPS maps completed at the 16–17-min mark. This meant processing times for RT-based maps generally required 12 to 13 min to complete, whereas TPS processing generally concluded within 2 min.

4 Discussion

4.1 Appraisal of the logger calibration procedure

The accuracy of the calibration procedure showed a high degree of reliability. Taken as a whole, the RMSE of 1.31 °C (i.e. when averaged across the seasons in Table 1) was the same as the RMSE produced in Webb et al. (2016), which also averaged 1.31 °C. This indicated that the procedure adopted in this study was suited to calibrating short-term

logger data for the extent of state. As demonstrated in Webb et al. (2016), the DEM is able to account for much of the temperature variability associated with elevation when forming the BoM grids, thereby allowing the calibration procedure to produce stable estimates. A closer inspection revealed that the DEM alone was able to achieve an average R^2 of 0.57 when used as the only guiding variable to de-trend the T_a observations in the calibration period. Given that increasing elevation strongly correlates to decreasing air temperature, the DEM was effective in accounting for the lapse rate, as expected (Hutchinson 1991). However, in this study, the BoM grids were formed via multiple regression models using the DEM along with covariates including SAGA wetness index (SWI), latitude (LAT), longitude (LONG) and distance to sea (DTS). This resulted in a larger proportion of the variability being modelled (average R^2 of 0.76). As demonstrated in Jarvis and Stuart (2001a), the covariates of “Northing” (similar to LAT), “distance to the east coast” (analogous to LONG), and distance to sea indices (similar to DTS) were important covariates for explaining daily

Table 6 Frequency of completion times for imported live BoM observations and corresponding T_a maps according to minutes from the nearest recording hour in the 7-day period from 7 to 13 February 2020

Minutes from recording hour	Frequency of BoM imports completed	Frequency of T_a maps completed from RT method	Frequency of T_a maps completed from TPS method
5	18	nil	nil
6	nil	nil	10*
7	nil	nil	8*
15	135	nil	nil
16	nil	nil	46**
17	nil	16*	89**
18	nil	2*	nil
20	1	nil	nil
21	nil	nil	1○
27	nil	132**	nil
28	nil	3**	nil
29	14	nil	nil
30	nil	nil	2○○
31	nil	nil	12○○
32	nil	1○	nil
41	nil	11○○	nil
42	nil	3○○	nil

* relates to BoM imports completed at 5 min; ** relates to BoM imports completed at 15 min; ○ relates to BoM imports completed at 20 min; ○○ relates to BoM imports completed at 29 min. “nil” means no value. TPS interpolation times were standardised to the RT interpolation times to allow comparisons

maximum and minimum temperatures. In this study, they were equally important because of their ability to account for the oceanic and latitudinal influences on air temperature (Legates and Willmott 1990; Wang et al. 2011). The SWI was also found to be an important predictor because of its ability to determine areas prone to cold air ponding and frost (Webb et al. 2018). In regards to the simple kriging of the model residuals, it was found that the calibration error was reduced by an average RMSE of 0.21 °C when compared to calibration without using geostatistical modelling (i.e. BoM grids formed without simple kriging). This indicated a degree of spatial dependence between BoM station locations, consistent in studies by Jarvis and Stuart (2001b) and Webb et al. (2016). When looking at the calibration procedure for each season, it was found that winter and spring produced predictions that were slightly more accurate compared to autumn and summer (Table 1). Of note, was the comparatively low R^2 value for winter (0.88), which would normally translate to a corresponding increase in prediction error. However, winter had superior MAE and RMSE values, especially when compared to summer. When analysing the prediction distributions, the calibrated predictions for summer were spread over a wider

range (average values of 3.63 °C and 33.79 °C for the 0 and 100th percentile, respectively) compared to estimates made for winter, which were narrower (average values of −3.24 °C and 14.68 °C for the 0 and 100th percentiles, respectively). This resulted in lower MAE and RMSE values because the potential error was confined to a narrower prediction range despite the lower R^2 . The comparatively higher standard deviation value for MAE suggested that the prediction errors were in fact slightly more erroneous for some of the calibrated loggers in winter.

Overall, it was found that summer tended to be more prone to greater error compared to the other seasons, particularly when measured against landform type. This was generally the case for loggers located on mountaintops/high ridges (MT) and upper slope/mesas (US) (Table 2). As explained in Mahrt (2006), predicting the variation of surface air temperature with surface elevation in areas of complex terrain depends on the interaction between three-dimensional topography, vegetation, soil characteristics, net radiation and speed of the large-scale flow. Moreover, the daytime spatial distribution of surface air temperature is highly influenced by solar irradiance, particularly in summer, which is governed directly by the three-dimensionality of the topography with respect to the solar zenith and azimuthal angles. Given MT sites would likely receive the highest daily total of net radiation during the day compared to other sites (since they mostly have an unobstructed view of the sky), combined with the spatial variations of vegetation and soil moisture, this can lead to dramatic variations of the daytime surface energy budget and heating of the atmosphere (Whiteman et al. 1989). As shown in Fig. 4, this had likely amplified prediction error during late mornings due to the inability of the calibration method to account for these factors on an hourly basis when solar irradiance was peaking. It should also be noted that the calibration procedure was also prone to error during the night reflecting an inability of the calibration model to account for heat loss via radiative cooling. This is likely exacerbated due to greater diurnal temperature fluctuations during summer which is likely to vary between sites in complex terrain areas (Minder et al. 2010). Given that the standard deviation (sd) for MT concerning MAE and RMSE values were 0.44 and 0.52, respectively and considerably higher than the sd values exhibited by the other landform types (Table 2), it demonstrates that diurnal variation between MT logger sites (and to a lesser degree US logger sites) may have been encountered in this study. Although, it should be noted that only a relatively small portion of loggers (4) were situated in MT areas and would therefore warrant a higher sampling density to draw a more informed conclusion. Nevertheless, an appropriate diurnal variation model could benefit future analysis, similar to that used in Parton and Logan (1981) and Zhu et al. (2013), or alternatively, the calibration procedure derived separately for night and day, as demonstrated in Lazzarini et al. (2014).

4.2 Appraisal of the RT and TPS interpolation methods

The RT and TPS interpolation over the whole of Tasmania was found to give reliable estimates of T_a . On average, the RMSE was 1.37 °C and 1.38 °C, respectively (i.e. when averaged across the seasons in Table 3), a difference of 0.06 °C and 0.07 °C, respectively, when compared to the RMSE exhibited by the calibration procedure (Table 1). This indicated that both interpolation procedures did not significantly degrade the T_a estimates when mapped across the state. When looking at variable importance used in the RT modelling, in terms of model usage frequency (Table 7), the DEM, LONG, LAT, SWI and DTS were identified as the five most utilised covariates. Collectively, these covariates were used in 8311 out of the 8760 models established, contributing most to each subsequent RT model formed. When these covariates were used solely as the guiding variables in the RT modelling, RMSE only increased slightly to 1.38 °C (i.e. when averaged across the seasons) reinforcing the importance of the variables for delineating T_a . For the TPS interpolation, this used LONG and LAT covariates as the two independent spline variables in addition to the DEM as the third independent variable representing the linear dependence on elevation. The relative success of the TPS method, particularly in summer and spring, reinforces these covariates importance for T_a modelling. Specifically, the dependence on LONG and LAT allows broad trends with respect to the location to be readily incorporated along with the DEM to account for the temperature lapse rates associated with Tasmanian topography (Hutchinson 1991). This tended to produce more accurate results in areas of

complex terrain such as in MT and US areas suggesting a propensity for the TPS method to better account for spatial variability associated with peaking solar irradiance, particularly in the middle part of the day. Conversely, the heightened temperature fluctuations in warmer months due to the lapse rate varying markedly over space and time (Minder et al. 2010; Whiteman et al. 1989) as well as the factors mentioned previously made it difficult for RT to account for this. As a consequence, this can lead to overfitting of data particularly with respect to the LONG and LAT resulting in unrealistic artefacts appearing in some maps, a phenomenon previously documented when mapping soil organic matter content (Pouladi et al. 2019). Nevertheless, in terms of predicting T_a during winter, spring and autumn nights and early mornings, the RT method was comparatively superior because of its ability to better model cold air drainage and frost conditions (Webb et al. 2016; Webb et al. 2018). Along with accounting for the complex interactions of inversion layers and lapse rates which can be non-linear (Noi et al. 2017; Xu et al. 2018). TPS fitted surfaces, on the other hand, tend to exhibit larger errors in winter due to insufficiently accounting for these conditions since inversion processes can disrupt the lapse rate relationship for which the algorithm is reliant (Hutchinson 1991; Jobst et al. 2017).

To benchmark the performance of the interpolation procedures, the current RT and TPS methods were compared with current methods of deriving T_a . The Scientific Information for Land Owners (SILO) database produces spatial grid estimates of daily minimum ($T_{\min}$) and maximum ($T_{\max}$) temperatures across Australia using trivariate TPS with interpolated products presented at 0.05° (~ 5 km) grid resolution (Jeffrey et al. 2001). The estimates are produced in a similar manner to the TPS predictions, albeit at a coarser spatial (5 km) and temporal (daily) resolution. The SILO predictions are readily accessible via the web (www.longpaddock.qld.gov.au/silo) and currently used as inputs to the Australian CliMate App (Australian CliMate Development Team 2016) to monitor crop growing degree days across the growing season (using updated daily inputs of $T_{\max}$ and $T_{\min}$). To compare methods, the calibration and interpolation procedures was readjusted to produce daily estimates of $T_{\max}$ and $T_{\min}$ in the evaluation period (instead of hourly T_a) with the interpolated predictions of RT and TPS cross-validated against actual $T_{\max}$ and $T_{\min}$ at the logger sites. Similarly, $T_{\max}$ and $T_{\min}$ were also extracted from the SILO database at the logger site locations and assessed against actual $T_{\max}$ and $T_{\min}$ for the same period. The comparison showed that both interpolations methods produced more accurate predictions of $T_{\max}$ and $T_{\min}$ than predictions provided by the SILO database (Table 8). RMSE for $T_{\max}$ interpolated via RT and TPS was lower by 0.04 °C and 0.11 °C, respectively, compared to the RMSE derived from SILO (i.e. 1.23 °C versus 1.19 °C and 1.12 °C). This was in addition to an improvement of 0.19 °C and 0.06 °C, respectively for the RT and TPS estimates of $T_{\min}$ registering 1.5 °C and 1.63 °C, compared to the SILO $T_{\min}$ RMSE of 1.69 °C.

Table 7 Frequency of covariates identified as contributing the highest proportion to model usage in each RT model formed for each hour ($n = 8760$) of the evaluation period

Covariate	Frequency
Elevation (DEM)	3829
Longitude (LONG)	1633
Latitude (LAT)	1286
Distance to sea (DTS)	910
SAGA wetness index (SWI)	653
Vertical distance to channel network	235
Diffuse insolation (summer)	92
Effective air flow heights	56
Wind direction (annual average)	27
Diffuse insolation (winter)	19
Wind effect (windward)	7
Elevation above channel network	5
Wind effect (leeward)	5
Wind speed (annual average)	3

Table 8 Validation statistics comparing RT and TPS derived estimates to SILO database estimates for $T_{\max}$ and $T_{\min}$ showing the coefficient of determination (R^2), concordance coefficient (P_c), mean absolute error (MAE, °C) and root-mean-square error (RMSE, °C) averaged for each logger site across the evaluation period

	RT		TPS		SILO	
	$T_{\max}$	$T_{\min}$	$T_{\max}$	$T_{\min}$	$T_{\max}$	$T_{\min}$
R^2						
mean	0.95	0.9	0.96	0.9	0.97	0.9
min	0.84	0.73	0.83	0.66	0.82	0.7
max	0.99	0.97	0.99	0.98	0.99	0.98
sd	0.02	0.04	0.02	0.05	0.02	0.05
P_c						
mean	0.97	0.94	0.97	0.93	0.96	0.92
min	0.87	0.78	0.87	0.67	0.66	0.67
max	0.99	0.98	0.99	0.98	0.99	0.99
sd	0.02	0.03	0.02	0.04	0.04	0.05
MAE						
mean	0.93	1.18	0.87	1.27	1.01	1.32
min	0.54	0.57	0.42	0.58	0.41	0.41
max	2.64	2.53	2.59	3.14	4.35	3.57
sd	0.27	0.31	0.27	0.43	0.56	0.52
RMSE						
mean	1.19	1.5	1.12	1.63	1.23	1.69
min	0.69	0.71	0.55	0.79	0.55	0.56
max	2.84	3.28	2.83	4.02	4.72	4.32
sd	0.33	0.38	0.33	0.53	0.59	0.61

sd, standard deviation; *min*, minimum; *max*, maximum

Another way to evaluate the current methodology was to assess predictions against the current arrangement of telemetered BoM stations. A common practice for many viticulturists in the Tasmanian wine industry is to garner current T_a observations from the nearest located station to monitor Growing Degree Day accumulation throughout the growing season (Smart, P 2018, pers. comm., 1 March 2018). In this context, each logger site was compared to their nearest located BoM station site to simulate where a location would derive T_a observations directly from the nearest BoM weather station, i.e. similar to what would occur for many established vineyard sites. The derived BoM observations (hourly) were then compared against the actual recordings provided by the associated logger site within the evaluation period allowing comparisons to be made with the current methodology. As expected, the method of sourcing observations from the closest BoM station yielded RMSE's of 2.23 °C, 2.34 °C, 2.23 °C and 2.25 °C for summer, autumn, winter and spring, respectively. This was markedly higher than RMSE values produced by the RT method with the RMSE lower by 0.84 °C, 0.95 °C, 0.89 °C and 0.89 °C and lower for the TPS method by 0.88 °C, 0.92 °C, 0.83 °C and 0.91 °C for summer, autumn, winter

and spring, respectively (Table 3). This confirmed that the current methodology, regardless of interpolation method was optimal in terms of producing reliable T_a estimates compared to observations sourced directly from a nearby public weather station site.

It should be commented that the TPS predictions made in summer and autumn was found to improve the predictions relative to the calibration procedure, particularly for landforms concerning MR, MT, OS and US. This suggested that the contributing logger sites to the TPS interpolation were less effective than anticipated. To test this, calibrated T_a estimates produced for the logger sites were removed from interpolation and revaluated using the BoM recordings as input to the modelling only (Table 9). This showed that the MAE for summer regarding the TPS was slightly improved by 0.01 °C compared to the validation statistics for the calibration procedures (Table 1), in addition to equalling the MAE for interpolation carried out using the original method (Table 3). Further still, the MAE across all seasons deviated by no less than 0.01 °C when compared to interpolation using BoM data only (Table 9) versus those carried out with all logger sites

Table 9 Validation statistics for RT and TPS interpolation using BoM recordings as the input data only (without calibrated logger site data) showing the coefficient of determination (R^2), concordance coefficient (P_c), mean absolute error (MAE, °C) and root-mean-square error (RMSE, °C) averaged for each logger site according to the season

	Summer		Autumn		Winter		Spring	
	RT	TPS	RT	TPS	RT	TPS	RT	TPS
R^2								
mean	0.88	0.93	0.87	0.91	0.83	0.88	0.88	0.92
min	0.74	0.73	0.69	0.68	0.62	0.45	0.74	0.75
max	0.96	0.98	0.95	0.97	0.94	0.97	0.96	0.97
sd	0.04	0.04	0.04	0.05	0.06	0.06	0.04	0.04
P_c								
mean	0.92	0.95	0.91	0.93	0.92	0.95	0.92	0.94
min	0.75	0.82	0.76	0.7	0.75	0.82	0.79	0.79
max	0.97	0.99	0.97	0.98	0.97	0.99	0.98	0.98
sd	0.03	0.03	0.03	0.04	0.03	0.03	0.03	0.03
MAE								
mean	1.36	1.01	1.24	1.04	1.2	1.05	1.27	1.01
min	0.85	0.44	0.75	0.48	0.67	0.46	0.77	0.52
max	2.56	2.59	2.68	2.94	3.3	3.38	2.32	2.3
sd	0.27	0.28	0.29	0.32	0.34	0.36	0.27	0.28
RMSE								
mean	1.82	1.37	1.65	1.45	1.62	1.44	1.69	1.38
min	1.15	0.62	0.98	0.67	0.87	0.62	1.05	0.69
max	3.17	3.25	3.3	3.66	3.9	4.06	2.91	2.86
sd	0.36	0.38	0.4	0.46	0.45	0.48	0.36	0.39

sd, standard deviation; *min*, minimum; *max*, maximum

involved (Table 3). Thus, it can be deduced that the calibration procedure did not greatly enhance prediction accuracy when logger site estimates were subjected to TPS interpolation. On the other hand, the calibration procedure greatly benefited the RT method given the MAE was greater by at least 0.18 °C for interpolation carried out using BoM data only (Table 9) versus those when carried out with logger sites available (Table 3).

4.3 Appraisal of mapping T_a in near real time

The proposed methodology applied in real-time showed it was capable of producing high-resolution T_a maps within the hour of the observation being recorded. It also showed a high degree of temporal reliability with all possible T_a hours mapped over the 7-day period from 7 to 13 February 2020. The processing time for TPS based maps, however, were markedly shorter than the RT method typically taking around 2 min to produce a single output compared to 12 min. This was mainly due to the TPS method being restricted to using only the DEM covariate to help guide TPS predictions, in contrast to the RT method which can use up to 14 (as depicted in Table 7). Thus, the overhead for processing RT predictions can be considerable owing to the many data partitions and linear sub-models to arrive at a single case prediction (Quinlan 1986). From this perspective, the TPS method was more suited to the real-time application.

5 Conclusions

The methods described in this study were successful for operational real-time spatial monitoring of T_a at a high spatiotemporal resolution across the state of Tasmania. Assessment of the T_a estimates produced by the calibration procedure showed a high degree of reliability, albeit a larger error in summer for high ridges/mountain tops and upper slope/mesas, particularly during late evenings and early mornings. When the calibrated T_a estimates underwent interpolation, the error increased slightly indicating that the RT and TPS interpolation method was a reliable spatial predictor of the T_a estimates. Spatial prediction error, however, increased markedly during early afternoons in summer for the RT method, partly reflecting an inability of the algorithm to cater to increased solar irradiance and greater spatial variability in complex terrain environments. In comparison, the TPS method performed better for these parts but was less accurate during early mornings in winter due to the inability to properly account for cold air drainage. It should be noted that the calibration procedure did not greatly enhance the TPS interpolation method and was equally effective when carried out using recordings provided by the BoM stations only. From this perspective, future TPS interpolation should be conducted without logger site data estimates contributing to the interpolation process. Nevertheless, the methods trialled in this study

proved to be an improvement compared to other methods of deriving T_a predictions, such as those sourced from the SILO database or more commonly, when accessed directly from the nearest located BoM weather station site. When applied to the operational monitoring of T_a , the mapping system, both interpolation procedures were able to provide spatial outputs within all hours of the BoM observations being recorded. However, the TPS method was by far more efficient in producing outputs in a timelier manner compared to those derived from the RT method. As such, the TPS method would be highly suited for applications requiring near real-time monitoring of hourly T_a over large heterogeneous land areas at the local scale (i.e. at a spatial resolution of 80 m) given the speed and relative accuracy of the outputs produced.

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